

Analysis 2
Serie 9
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Volume

$$x = r \cos(\theta) \quad y = r \sin(\theta) \quad dA = r dr d\theta$$

$$x^2 + y^2 + z^2 = 8 \Rightarrow r^2 + z^2 = 8 \Rightarrow z = \sqrt{8 - r^2}, \quad z = \frac{r^2}{2}$$

$$r^2 + \left(\frac{r^2}{2}\right)^2 = 8 \Leftrightarrow r^4 + 4r^2 - 32 = 0 \Rightarrow r_{1,2}^2 = 4, -8 \Rightarrow 0 < r < 2$$

The formular is therefore:

$$\begin{aligned} \int_0^{2\pi} \int_0^2 \int_{\frac{r^2}{2}}^{\sqrt{8-r^2}} r dz dr d\theta &= 2\pi \int_0^2 r \sqrt{8-r^2} - \frac{r^3}{2} dr \\ &= 2\pi \left[-\frac{1}{3}(8-r^2)^{\frac{3}{2}} - \frac{r^4}{8} \right]_0^2 \\ &= \frac{2}{3}\pi(16\sqrt{2} - 8 - 6) \\ &= \frac{2\pi}{3}(16\sqrt{2} - 14) \end{aligned}$$

Multiple Intergals

$$1. \int_0^2 \int_0^1 (x^3 + 3x^2y + y^3) dy dx = \int_0^2 x^3 + \frac{3}{2}x^2 + \frac{1}{4} dx = \frac{16}{4} + \frac{8}{2} + \frac{1}{2} = \frac{17}{2}$$

$$2. \int_0^\pi \int_x^\pi x \cos(x+y) dy dx = \int_0^\pi x(\sin(x+\pi) - \sin(2x)) dx$$

$$\int_0^\pi x \sin(x) dx = [\sin(x) - x \cos(x)]_0^\pi = \pi$$

$$\int_0^\pi x \sin(2x) dx = \left[-\frac{1}{2}x \cos(2x) + \frac{1}{4} \sin(2x) \right]_0^\pi = -\frac{\pi}{2}$$

$$\Rightarrow V = \frac{\pi}{2}$$

$$3. \int_1^2 \int_1^{3-y} \frac{1}{(x+y)^3} dx dy =$$

$$\int_1^2 \int_{1+y}^3 u^{-3} du dy =$$

$$\int_1^2 \left[-\frac{1}{2}u^{-2} \right]_{1+y}^3 =$$

$$\int_1^2 \frac{1}{2(y+1)^2} - \frac{1}{18} dy =$$

$$\left[-\frac{y}{18} \right]_1^2 + \left[\frac{1}{2}u^{-1} \right]_2^3 = \frac{1}{12} - \frac{1}{18} = \frac{1}{36}$$

Fubini Theorem.

$$\begin{aligned} 1. \quad \int_0^{\sqrt{\pi}} \int_x^{\sqrt{\pi}} \sin(y^2) dy dx &= \int_0^{\sqrt{\pi}} \int_0^y \sin(y^2) dx dy \\ &= \int_0^{\sqrt{\pi}} y \sin(y^2) dy \\ &= \int_0^{\pi} \frac{\sin(u)}{2} du = \frac{1}{2}(1 + 1) = 1 \end{aligned}$$

2.

$$\begin{aligned} \int_{-1}^1 \int_{|y|}^1 (x+y)^2 dx dy &= \int_0^1 \int_{-x}^x (x+y)^2 dy dx \\ &= \int_0^1 \left[x^2 y + xy^2 + \frac{1}{3} y^3 \right]_{-x}^x dx \\ &= \int_0^1 2x^3 + 0 + \frac{2}{3} x^3 dx \\ &= \frac{2}{3} \end{aligned}$$

Counterexample to Fubini.

Let $f : [0, \infty) \rightarrow \mathbb{R}$, defined by

$$f(x, y) = \begin{cases} e^{y-x} & x > y \geq 0 \\ -e^{x-y} & 0 \leq x < y \end{cases}$$

1.

$$\begin{aligned} \int_0^\infty \left\{ \int_0^\infty f(x, y) dx \right\} dy &= \int_0^\infty \left\{ \int_0^y -e^{x-y} dx + \int_y^\infty e^{y-x} dx \right\} dy \\ &= \int_0^\infty \{e^{-y} - 1 + 1\} dy = 1 \end{aligned}$$

2

$$\begin{aligned} \int_0^\infty \left\{ \int_0^\infty f(x, y) dy \right\} dx &= \int_0^\infty \left\{ \int_0^x -e^{y-x} dy + \int_x^\infty e^{x-y} dy \right\} dx \\ &= \int_0^\infty \{-e^{-x} - 1 + 1\} dx = -1 \end{aligned}$$

Volume of the cone over a set.

Fix $t \in [0, 1]$. For this t the set becomes

$$C\Omega_t = \{x \in \mathbb{R}^n : x \in (1-t)\Omega\} = (1-t)\Omega$$

Using homogeneity of μ_n

$$\mu_n((1-t)\Omega) = (1-t)^n \mu_n(\Omega)$$

Since $0 \leq t \leq 1$ we can integrate over t to get the full set

$$\mu_{n+1}(C\Omega) = \int_0^1 (1-t)^n \mu_n(\Omega) = \mu_n(\Omega) \int_0^1 (1-t)^n = \frac{\mu_n(\Omega)}{1+n}$$

□

We want to show that

$$\mu_n(T_n) = \mu_n(\{a_1 e_1 + \dots + a_n e_n : 0 \leq a_i \leq 1, a_1 + \dots + a_n \leq 1\}) = \frac{1}{n!}$$

Proof by induction

Base case $n = 1$

$$\mu_1(T_1) = \mu_1(\{a_1 e_1 : 0 \leq a_1 \leq 1\}) = \int_0^1 dx = 1$$

Taking the cone is exactly the same as going one dimension higher.

$$\mu_{n+1}(CT_n) = \mu_{n+1}(T_{n+1}) = \frac{\mu_n(T_n)}{n+1}$$

Therefore

$$\mu_n(T_n) = \frac{1}{n!}, \quad \forall n \geq 1$$

□

Gaussian integrals.

Let $n \in \mathbb{N}$ and $A \in M_{n \times n}(\mathbb{R})$ be a positive definite matrix. Show that

$$\int_{\mathbb{R}^n} e^{-\langle Ax, x \rangle} dx = \frac{\pi^{\frac{n}{2}}}{\sqrt{\det(A)}}$$

Since A is positive definite and $x \in \mathbb{R}^2$. $\exists Q \in M_{n \times n}(\mathbb{R})$ such that

$$A = Q\Lambda Q^T$$

Where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$

$$\Rightarrow \langle Ax, x \rangle = x^T Ax = x^T Q\Lambda Q^T x = (Qx^T)^T \Lambda (Qx^T) = y^T \Lambda y = \sum_{i=1}^n \lambda_i y_i^2$$

$y = Q^T x \in \mathbb{R}^n$. Then, by the change of variables formula,

$$dx = |\det(Q)| dy.$$

Since Q is orthogonal, we have $|\det(Q)| = 1$. Hence,

$$dx = dy$$

and

$$\det(A) = \det(Q\Lambda Q^T) \det(Q) \det(\Lambda) \det(Q^T) = \prod_{i=1}^n \lambda_i$$

Therefore

$$\begin{aligned}\int_{\mathbb{R}^n} e^{-\langle Ax, x \rangle} dx &= \int_{\mathbb{R}^n} e^{-\sum_{i=1}^n \lambda_i y_i^2} dx \\ &= \prod_{i=1}^n \int_{\mathbb{R}} e^{-\lambda_i y_i^2} dy_i \\ &= \prod_{i=1}^n \sqrt{\frac{\pi}{\lambda_i}} = \frac{\pi^{\frac{n}{2}}}{\sqrt{\det(A)}}\end{aligned}$$

□